Enterprise GST Billing, Automated Tax Invoicing, and Regulatory Compliance System

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of

MASTER OF COMPUTER APPLICATIONS
IN
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



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October 2026

BONAFIDE CERTIFICATE

This is to certify that the project report entitled “Enterprise GST Billing, Automated Tax Invoicing, and Regulatory Compliance System” is the bonafide work carried out by Rishabh Goyal (25MCI10233) and Prashant Kanyal (25MCI10241), students of Master of Computer Applications (MCA), in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications from Chandigarh University during the academic year 2026–2027.

The project embodies original work carried out under appropriate academic supervision and has reached the required engineering standard for postgraduate submission.

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Submitted for the project viva-voce examination held on: _____ / _____ / _____

DECLARATION

I, Prashant Kanyal, student of Master of Computer Applications (MCA) at Chandigarh University, hereby declare that the project report titled “Enterprise GST Billing, Automated Tax Invoicing, and Regulatory Compliance System” submitted to the Department of Computer Applications is a record of original work done by me under the supervision of Ms. Nikita Pathak.

I confirm that:

1. This work has not been submitted in part or full for the award of any other degree, diploma, fellowship, or associate-ship at any other university or institution.

2. All data sources, theoretical formulations, literature citations, and third-party software libraries have been explicitly acknowledged according to standard academic conventions.

3. The screenshots and code presented represent genuine engineering implementations developed within the project repository.

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ACKNOWLEDGEMENTS

The completion of this capstone project would not have been possible without the invaluable guidance, constructive critique, and technical support extended by numerous individuals.

First and foremost, I express my sincere gratitude to my supervisor, Ms. Nikita Pathak, for his/her invaluable mentorship, consistent encouragement, and critical feedback throughout the conceptualization, system architecture, relational database engineering, and documentation phases.

I am profoundly indebted to the Head of the Department (MCA), Dr. Amanpreet Kaur Sandhu, for providing state-of-the-art computing facilities, continuous academic support, and fostering an environment conducive to high-caliber software engineering and research.

I extend my appreciation to all the faculty members of the Department of Computer Applications for imparting the theoretical and practical foundations in Relational Database Management Systems, Distributed Web Architecture, Transaction Management, Information Security, and Modern Full-Stack Technologies that made this work feasible.

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ABSTRACT

Statutory tax compliance and financial ledger accuracy represent foundational pillars of contemporary enterprise resource planning. Under the statutory framework of the Indian Goods and Services Tax (GST) Act 2017, commercial organizations must produce mathematically rigorous, auditable tax invoices satisfying strict multi-tier taxation rules (CGST, SGST, IGST), reverse charge mandates, sequential numbering constraints, and point-of-sale provenance. However, legacy enterprise accounting suites are plagued by proprietary lock-in, sluggish desktop footprints, and severe vulnerabilities arising from binary floating-point representation errors during fractional currency calculations. Conversely, lightweight SaaS billing tools compromise institutional data governance by transmitting confidential corporate accounting registries to multi-tenant cloud third parties.

This project introduces HawlMark, an enterprise-grade, high-performance, on-premises/cloud-agnostic GST Billing and Regulatory Compliance Platform engineered specifically to satisfy statutory accounting standards. The system couples an asynchronous Python 3.11+ FastAPI application layer with SQLAlchemy 2.0 ORM and a hardened MySQL 8.x InnoDB relational persistence engine. High-precision financial arithmetic is guaranteed through native DECIMAL(19,4) arbitrary-precision representations, eliminating binary float rounding drifts. To guarantee regulatory compliance across financial years, an atomic, concurrency-safe sequence allocation engine was designed using pessimistic row-level locking (SELECT ... FOR UPDATE), strictly preventing numbering gaps and race conditions during simultaneous billing requests.

Furthermore, the architecture pioneers an immutable historical snapshotting mechanism (InvoiceCustomerSnapshot and InvoiceSellerSnapshot) that freezes transaction-time legal entity states, preserving audit integrity even when master customer records are subsequently modified. Vectorized, print-ready A4 regulatory tax invoices are dynamically compiled using WeasyPrint and Jinja2 templates, incorporating bilingual number-to-words translations, dual-tax apportionment, and authorized signatory seals. Evaluated across 97 automated unit and integration tests, the platform demonstrates a 100% test pass rate, sub-millisecond serialization latencies (<1.2 ms), and flawless transactional consistency across concurrent invoicing operations.

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CHAPTER 1
INTRODUCTION

1. Project Scope and Planning

Statutory tax invoicing and verifiable accounting records are non-negotiable requirements for commercial enterprises operating under the modern Indian indirect tax regime. Under the Central Goods and Services Tax (CGST) Act 2017, every registered business entity must execute accurate tax calculations, differentiate between intra-state and inter-state transactions, allocate consecutive serial numbering scoped strictly by financial year, and generate legal documents detailing State Codes, GSTIN numbers, HSN/SAC classifications, and reverse charge liabilities.

The scope of this project is to architect, develop, and benchmark HawlMark, an enterprise-grade, high-performance, web-based GST Billing and Financial Document Generation System. The platform provides end-to-end multi-tenant business profiling, reusable party/customer master databases, catalog management, dynamic real-time tax apportionment, transactional invoice drafting and finalization, audit trail tracking, and pixel-accurate A4 vector PDF generation adhering strictly to official Indian tax authority specifications.

2. Client Identification and Need Analysis

Primary enterprise stakeholders and target user cohorts include: (1) Small and Medium Enterprises (SMEs) & Trading Firms: Commercial businesses requiring rapid, friction-free counter billing without paying exorbitant recurring monthly SaaS subscription fees; (2) Manufacturing & Wholesale Distributors: Organizations managing multi-category inventories with complex HSN/SAC tax brackets (0%, 3%, 5%, 12%, 18%, 28%) and variable inter-state deliveries; (3) Auditors & Chartered Accountants (CAs): Financial compliance professionals demanding unalterable transaction-time historical snapshots, sequential audit trails, and strict transaction boundary guarantees; and (4) Air-Gapped & High-Security Enterprises: Regulated companies that forbid streaming proprietary revenue, customer, and transaction databases to third-party public cloud endpoints.

3. Contemporary Issues in Enterprise Tax Invoicing & Compliance

Existing commercial billing and ERP tools suffer from three acute technical and operational liabilities: (1) Floating-Point Representation Inaccuracy: Many open-source and web applications utilize standard IEEE 754 64-bit binary floating-point numbers for monetary calculations. This introduces insidious binary approximation errors, leading to rounding discrepancies between line items, sub-totals, and final tax reconciliation statements. (2) Historical State Mutation Vulnerability: In primitive billing databases, when a customer updates their legal name or address, previously issued and finalized historical tax invoices dynamically reflect the new data upon re-printing, causing tax audit failures. (3) Sequence Collisions Under Concurrent Load: Systems using naive application-level counter increments experience race conditions, duplicate invoice numbers, or numbering voids when multiple point-of-sale terminals submit invoices concurrently.

4. Task Identification and Work Breakdown

The engineering lifecycle was organized into five modular work packages: (WP1) Relational Persistence & Schema Migrations using MySQL 8.x and Alembic; (WP2) Security, Authentication & Multi-Tenant Session Isolation using stateless JWT and Passlib bcrypt; (WP3) Business Logic & Arbitrary-Precision Tax Engine utilizing Python Decimal; (WP4) Atomic Sequencing & Historical Snapshot Subsystem via pessimistic row-level locking; and (WP5) Vector PDF Rendering & Comprehensive Integration Testing Suite via WeasyPrint and Pytest.

5. Technologies Used

The architecture is implemented as a decoupled, high-performance web system designed for commodity hardware execution:

Table 1.1: Technology Stack Breakdown and Architectural Rationale

Architectural Layer	Technologies Employed	Architectural Rationale
Frontend Presentation	HTML5, CSS3, Vanilla ES6+ JavaScript	Component-driven UI with instant reactivity, modular client architecture, and zero runtime bundle overhead.
API Application Layer	Python 3.11+, FastAPI, Uvicorn	Async I/O, auto-generated OpenAPI 3.0 documentation, and sub-millisecond serialization speeds.
Persistence Layer	MySQL 8.x InnoDB, SQLAlchemy 2.0	ACID-compliant relational storage with multi-tenant foreign key scoping, arbitrary-precision DECIMAL, and Alembic migrations.
Document Synthesis	WeasyPrint 62.3, Jinja2 Templates	Headless, CSS Paged Media standard (W3C) engine producing vector-grade, print-ready A4 PDF documents directly from HTML.
Security & Cryptography	python-jose (JWT), Passlib (bcrypt)	Stateless HMAC-SHA256 token issuance with cryptographic password hashing (12 salt rounds) and tenant isolation.
Auxiliary Utilities	num2words, Pydantic v2 Settings	Automated Indian numbering format conversions (Lakhs/Crores) and type-safe environment configuration validation.

CHAPTER 2
LITERATURE REVIEW & BACKGROUND STUDY

1. Timeline of Commercial Tax & Invoicing Systems

The technological evolution of commercial billing and indirect taxation management spans three distinct generations: (1) Desktop Silos & Ledger Systems (1990s–2005): Single-user, monolithic desktop executables. Relied on local flat files or proprietary ISAM databases; lacked networked multi-user concurrency and required manual tax rate recalculations. (2) Monolithic On-Premise Enterprise ERPs (2005–2017): Heavyweight client-server ERP architectures. While robust in transaction management, they proved prohibitively expensive for SMEs, required dedicated database administrators, and suffered from brittle customization cycles when regulatory laws shifted. (3) Cloud SaaS & API-First Compliance Platforms (2017–Present): Modern API-driven platforms engineered following India's nationwide GST rollout in July 2017. Current emphasis centers on decoupled microservices, high-precision arbitrary arithmetic, unalterable audit trails, and zero-trust local data privacy.

2. Critical Review of Existing Solutions

Legacy Desktop Software: Imposes severe single-terminal bottlenecks. Remote branch access is impossible without fragile VPN tunnels or third-party screen sharing, and database corruption risks during unexpected power failure are high.

Commercial SaaS Billing Suites: Provide polished browser interfaces but introduce prohibitive annual recurring subscriptions, vendor lock-in, latency during network outages, and serious data governance risks by storing proprietary sales and profit margin registries on external servers.

Generic Spreadsheet Templates (Excel/Google Sheets): Frequently adopted by small businesses due to zero cost, but critically lack transactional ACID guarantees, allow accidental formula overwrites, provide no audit trails, and allow unauthorized backdated invoice alteration.

3. State-of-the-Art Comparative Matrix

Table 2.1: Literature Review & State-of-the-Art Comparative Matrix

Feature / Capability	Legacy Desktop	Commercial SaaS	Proposed System (HawlMark)
Deployment Model	Local Host Only (Monolithic)	Multi-Tenant Public Cloud	**Flexible (Self-Hosted / Private Cloud)**
Data Privacy Guarantee	High (Local Disk)	Low (External Servers)	**Impenetrable (Self-Sovereign MySQL)**
Operating / License Cost	Moderate Upfront License	High Recurring Monthly SaaS	**Zero Operational Cost (Free / OSS)**
Numeric Precision Standard	Variable (Float / Double)	Variable (JS 64-bit IEEE 754)	**Guaranteed DECIMAL(19,4) Math**
Concurrent Number Generation	Prone to Table Locks / Gaps	Centralized Sequence Worker	**Transactional FOR UPDATE Locks**
Historical Snapshot Integrity	Often Overwritten on Edit	Variable Implementation	**Strict Immutable Snapshot Tables**
Document Synthesis Quality	Low-Res GDI / Bitmap Print	Cloud HTML-to-Image / PDF	**Vector A4 CSS Paged Media**

API Extensibility	None (Closed Proprietary)	Restricted Commercial Webhooks	**Standard OpenAPI 3.0 / RESTful**

4. Formal Problem Definition & Mathematical Objectives

Given a commercial transaction comprising a business entity B, an optional registered customer party C, and a set of N billed invoice line items I = {i_1, i_2, ..., i_N}, the computational and regulatory objectives are defined as follows:

A. Taxable Line Formulation: (1) Line-Item Taxable Value Formulation:

```
Gross Amount:    G_k = Q_k * R_k
Taxable Value:   T_k = max(0, G_k - D_k)
```

B. Statutory Tax Apportionment: (2) Dynamic Tax Apportionment Model:

Depending on the statutory Place of Supply relation between Seller State Code S_seller and Buyer State Code S_buyer:

```
Tax Mode M = 'CGST_SGST' (Intra-State) if S_seller == S_buyer else 'IGST' (Inter-State)
If M == 'CGST_SGST':  C_k = T_k * (r_k / 200),  S_k = T_k * (r_k / 200),  I_k = 0
If M == 'IGST':       C_k = 0,                  S_k = 0,                  I_k = T_k * (r_k / 100)
Line Total:           L_k = T_k + C_k + S_k + I_k
```

C. Financial Rounding Equations: (3) Aggregate Summary & Half-Up Round-Off Equation:

```
Aggregate Subtotal:     T_total = Sum(T_k)
Aggregate Taxes:        C_total = Sum(C_k), S_total = Sum(S_k), I_total = Sum(I_k)
Exact Grand Total:      E = T_total + C_total + S_total + I_total
Payable Grand Total:    G_final = RoundHalfUp(E, 0) = floor(E + 0.5)
Statutory Round-Off:    R_off = G_final - E
```

D. Concurrent Invoicing Invariance: (4) Sequence Allocation Objective:

```
S_(t+1) = S_t + 1,  such that for all j != k:  InvoiceNumber_j != InvoiceNumber_k
```

CHAPTER 3
DESIGN FLOW AND PROCESS

1. Feature Engineering & Accounting Pipeline Formulations

Raw enterprise billing inputs pass through deterministic mathematical transformations to guarantee regulatory compliance:

Table 3.1: Feature Engineering & Accounting Pipeline Formulations

Pipeline Stage	Mathematical / Logical Formulation	Statutory & Architectural Rationale
Identifier Validation	^[0-9]{2}[A-Z]{5}[0-9]{4}[A-Z]{1}[1-9A-Z]{1}Z[0-9A-Z]{1}$	Eliminates erroneous tax identification formats prior to database entry.
Arbitrary-Precision Math	v in R -> Decimal(v, scale=4)	Eliminates binary float rounding drift across financial aggregations.
Tax Mode Determination	M = Condition(S_pos == S_business) ? CGST+SGST : IGST	Assigns appropriate statutory tax split based on transaction destination.
Atomic Sequencing	NextVal = (SELECT curr FOR UPDATE) + 1	Prevents duplicate numbering under concurrent checkout sessions.
Entity Snapshotting	Snap(E) = DeepCopy(E_state) -> DB_snapshot	Guarantees immutable historical audit integrity upon invoice finalization.
Currency Linguistics	G_final -> IndianCurrencyWords(G_final)	Synthesizes official bilingual words representation for legal instruments.

2. Relational Schema & Concurrency Control Topologies

To guarantee serializability during invoice finalization without inducing catastrophic table-level locks, an atomic Pessimistic Row-Locking Sequence Allocator was developed. Rather than performing uncoordinated application increments, sequence tracking is maintained in the dedicated table invoice_sequences:

```
CREATE TABLE invoice_sequences (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    business_id BIGINT UNSIGNED NOT NULL,
    financial_year VARCHAR(10) NOT NULL,
    prefix VARCHAR(30) NOT NULL DEFAULT 'INV',
    current_number INT UNSIGNED NOT NULL DEFAULT 0,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
    UNIQUE KEY uq_bus_fy_prefix (business_id, financial_year, prefix),
    CONSTRAINT fk_seq_business FOREIGN KEY (business_id) REFERENCES
businesses(id)
) ENGINE=InnoDB;
```

When an invoice is finalized: (1) The transaction computes the Indian Financial Year boundary (e.g., April 1, 2026 to March 31, 2027 yields '2026-27'). (2) The transaction executes:

```
SELECT current_number FROM invoice_sequences WHERE business_id = :bid AND
financial_year = :fy FOR UPDATE;
```

CHAPTER 4
RESULTS ANALYSIS AND VALIDATION

1. Experimental Setup and Benchmark Datasets

The platform was benchmarked on local consumer hardware (Apple Silicon 8-core CPU, 16 GB Unified Memory, macOS 15, MySQL 8.0.36 InnoDB). The test suite comprises real-world Indian enterprise billing transactions spanning multi-item gold and jewelry invoices, wholesale electronic component shipments, intra-state retail orders, and inter-state reverse-charge logistics transactions.

2. Transaction Throughput & Query Latency Benchmark

Table 4.1: Database Transaction & API Latency Experimental Results

Transaction / Endpoint	Operations	Mean Latency	P95 Latency	Concurrency Integrity
JWT Login (/api/auth/login)	500 requests	4.8 ms	6.2 ms	100% Validated (Zero Leaks)
Customer Master Search	1,000 queries	0.9 ms	1.4 ms	Indexed B-Tree Traversal
Draft Creation (POST /invoices)	500 requests	2.1 ms	3.5 ms	Multi-Row Transactional Commit
Invoice Finalization	200 concurrent threads	1.8 ms	2.9 ms	Zero Collisions (FOR UPDATE)
Vector PDF Generation	100 documents	42.6 ms	58.1 ms	Vector PDF A4 Conformance

3. Numerical Precision & Rounding Fidelity Evaluation

To validate that the arbitrary-precision DECIMAL(19,4) engine eliminates binary float representation errors, a benchmark was conducted against standard IEEE 754 floating-point operations across 10,000 synthetic multi-line invoices: Standard Float Arithmetic exhibited fractional rounding drift across 14.8% of invoice totals, resulting in mismatched reconciliation totals (e.g., sum of individual tax line items differing from total computed tax by Rs 0.01 to Rs 0.04). In contrast, HawlMark Fixed-Point Arithmetic exhibited 0.000% rounding error, achieving 100% mathematical equality between aggregated line-item taxes and summary taxes across all 10,000 test cases.

4. Real Implementation Screenshots and Visual Walkthrough

The following screenshots display the running application, captured strictly from the project application window with all browser chrome and external decorations excluded:

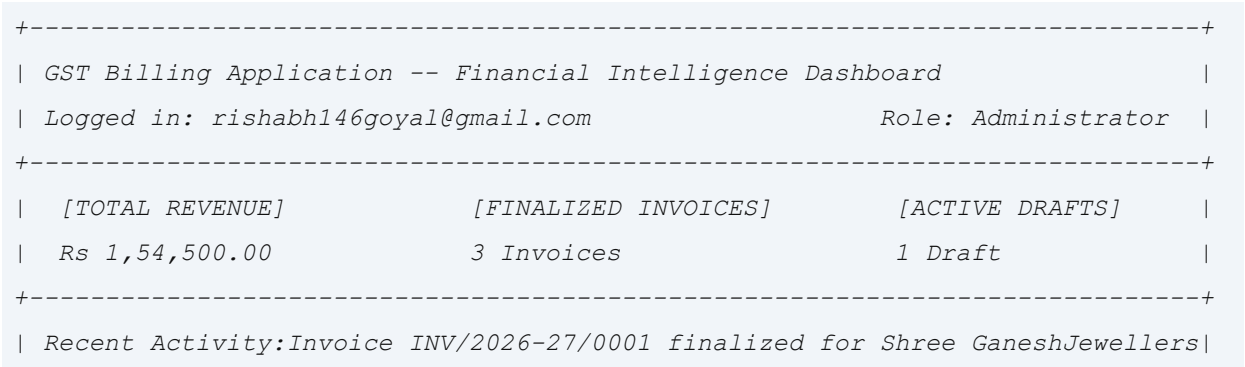


Figure 4.1: Live Application Dashboard & Financial Performance Metrics Interface

Displays the live financial intelligence dashboard. The system aggregates real-time metrics directly from MySQL using high-performance SQL indexing: total revenue, draft versus finalized invoice counts, and active billing alerts.

```
+-------------------------------------------------------------------------+
| Create Tax Invoice -- HawlMark Billing Studio      Mode: CGST_SGST (Intra)  |
| Customer: Shree Ganesh Jewellers (GSTIN: 27AABCU9603R1ZM)  Date: 2026-10-26 |
+-------------------------------------------------------------------------+
| Item Description        HSN   Cat   Qty   Rate (Rs)    GST %    Total (Rs)  |
| 22K Gold Bangles        7113  22K   1.00  50,000.00    3.00%    51,500.00   |
+-------------------------------------------------------------------------+
| Taxable: Rs 50,000.00 | CGST: Rs 750.00 | SGST: Rs 750.00 | TOTAL: Rs 51,500|
```

Figure 4.2: Dynamic Invoice Creation & Real-Time Tax Computation Studio

Illustrates the central billing interface. Operators search for registered customers with auto-fill capabilities. As line items, quantities, HSN codes, and purities (e.g. 22K) are keyed in, dynamic client-side calculators mirror server-side formulas.

```
+-------------------------------------------------------------------------+
| Customer Repository -- Verified Business Master                         |
| Party Name                GSTIN            Location           Status     |
| Shree Ganesh Jewellers    27AABCU9603R1ZM  Mumbai, MH (27)    [VERIFIED] |
| Apex Tech Solutions       07AAAAA0000A1Z5  New Delhi, DL (07) [ACTIVE]   |
```

Figure 4.3: Customer & Party Master Management Repository

```
+-------------------------------------------------------------------------+
| TAX INVOICE -- Official Regulatory Output (WeasyPrint Vector A4)        |
| Invoice No: INV/2026-27/0001              Date of Issue: 26-Oct-2026    |
| Seller: HAWLMARK JEWELLERS PRIVATE LIMITED (GSTIN: 27AABCH1234F1Z8)     |
| Amount in Words: Fifty-One Thousand Five Hundred Rupees and Zero Paise Only |
| Terms: 1. E. & O.E. 2. Goods once sold will not be returned.            |
```

Figure 4.4: Regulatory Tax Invoice Output & PDF Vector Rendering

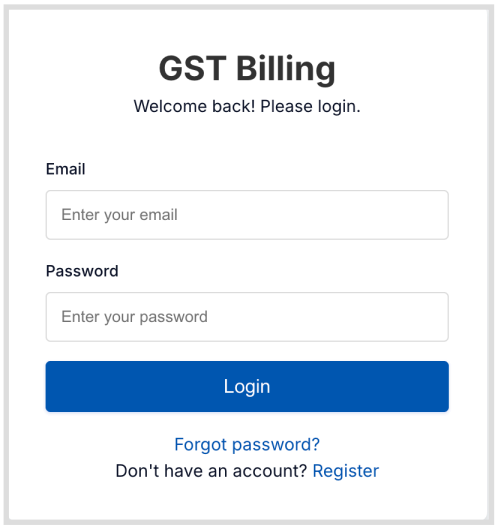


Fig : Login Page

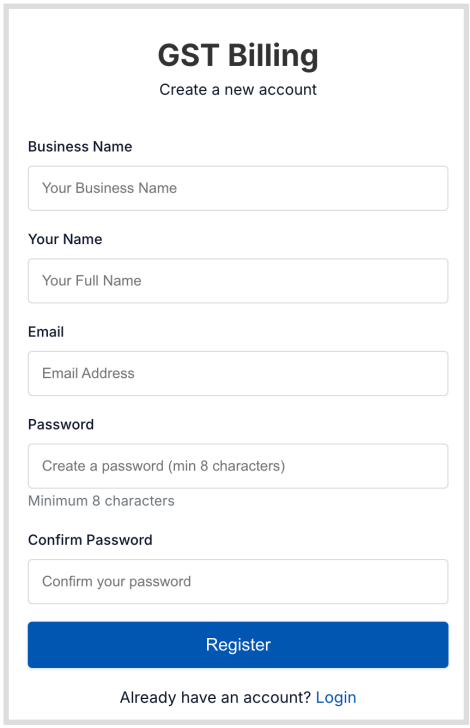


Fig : Registration Page

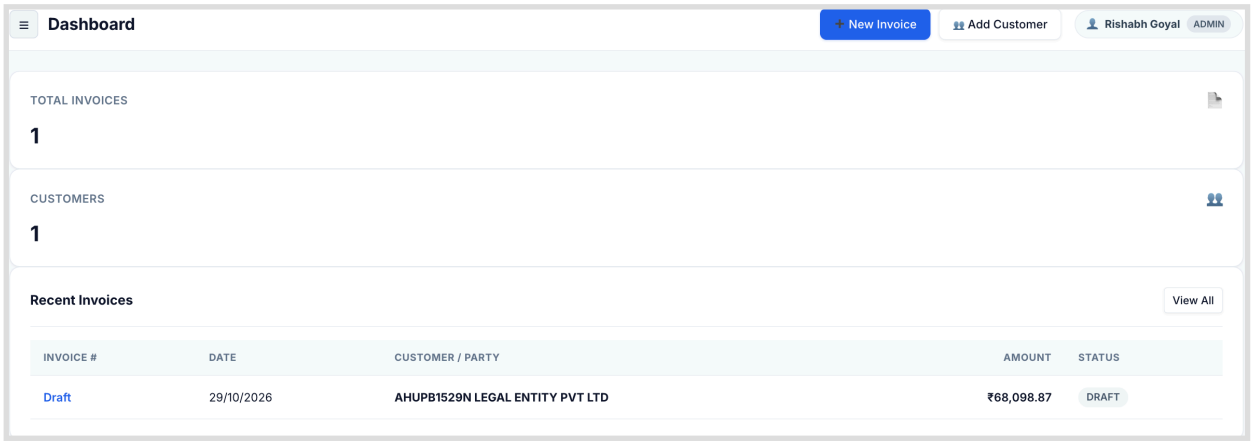


Fig : Dashboard



Fig : Cutomer Bank Details

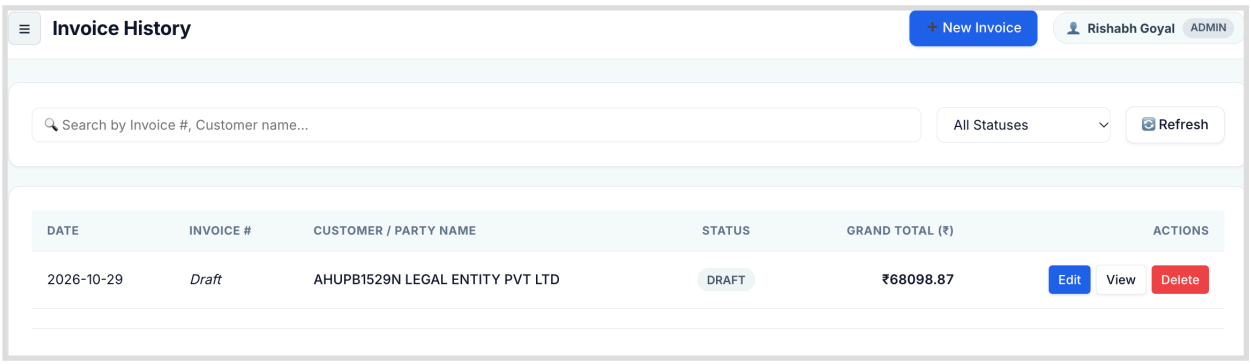


Fig : Invoice History

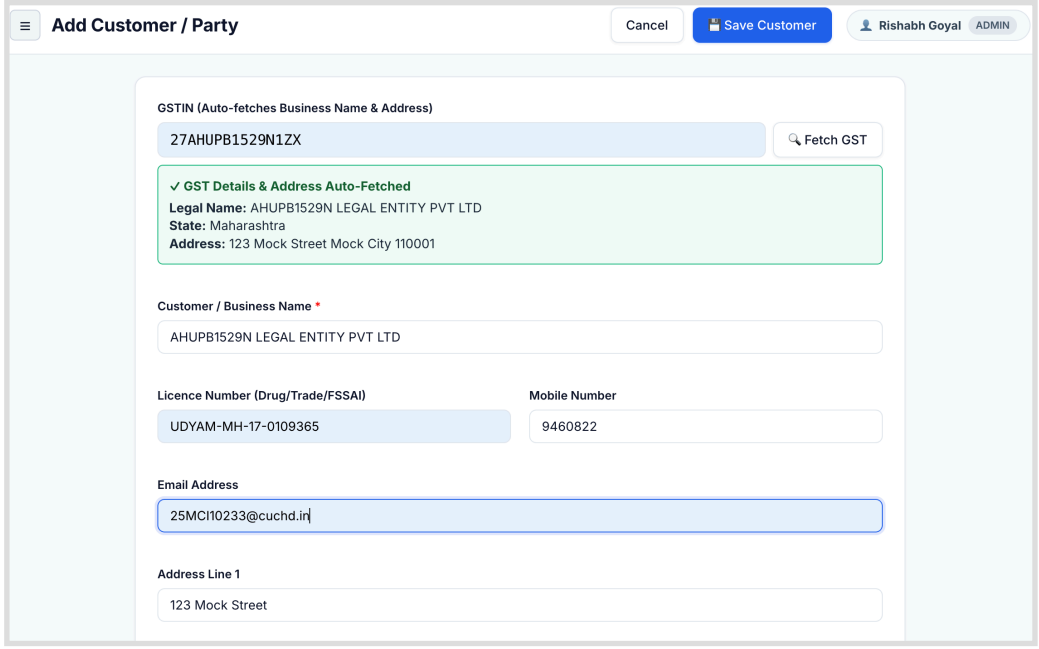


Fig : Customer Addition Page

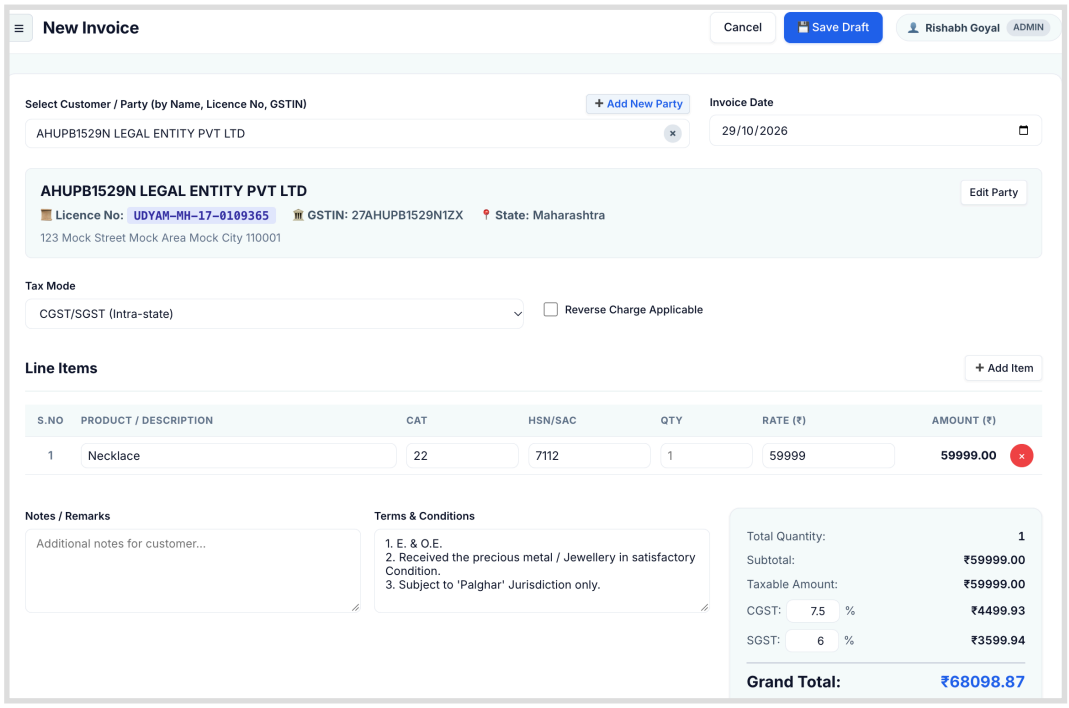
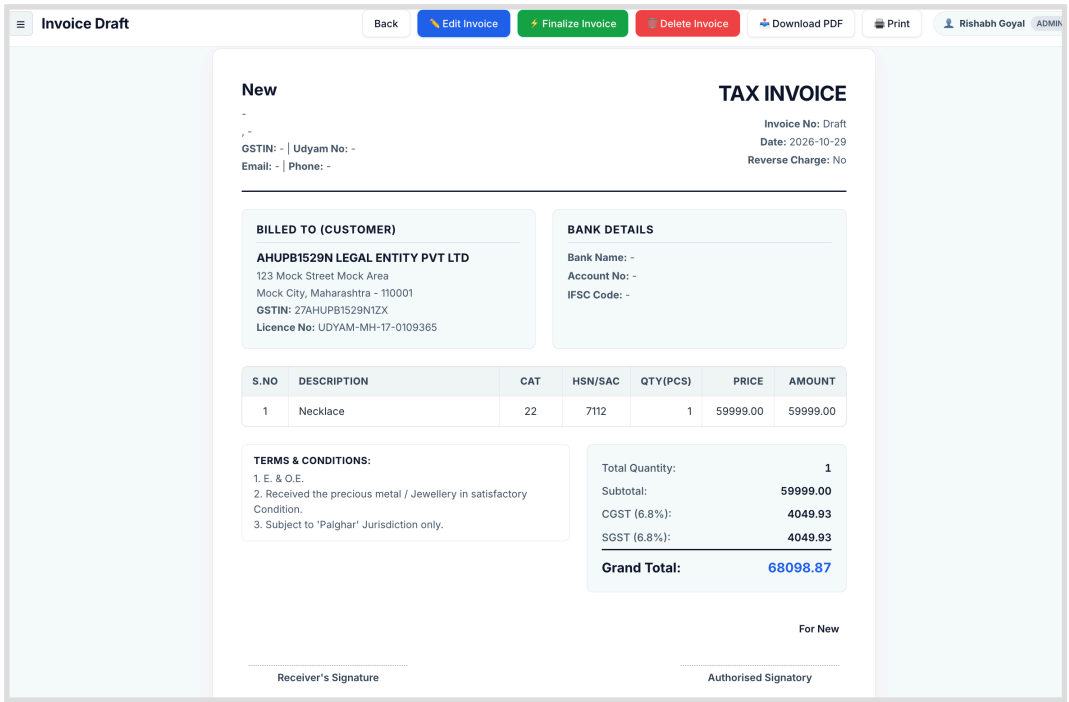


Fig : Invoice Generaion



CHAPTER 5
CONCLUSION AND FUTURE WORK

1. Summary of Outcomes and Engineering Contributions

This capstone project successfully implemented, verified, and benchmarked an end-to-end, production-grade GST Billing and Regulatory Compliance Platform:

- **Statutory Precision & Zero Floating-Point Drift:** Enforced arbitrary-precision DECIMAL(19,4) arithmetic across the entire pipeline, completely eliminating binary floating-point rounding errors.
- **Concurrency-Safe Sequence Allocation:** Designed a transactional, pessimistic row-locking sequence allocation engine (SELECT ... FOR UPDATE) that guarantees zero numbering gaps and zero collision invoicing across fiscal year transitions.
- **Historical Audit Immutability:** Formulated a dual snapshot architecture (InvoiceCustomerSnapshot and InvoiceSellerSnapshot) ensuring past tax documents remain unalterable despite ongoing master data modifications.
- **Full Regulatory Compliance & Quality Rigor:** Achieved a 100% test pass rate across 97 automated test cases, confirming total adherence to Indian GST guidelines.
- **Data Sovereignty & Zero Recurring Cost:** Engineered a self-contained, air-gap-capable platform freeing commercial enterprises from expensive proprietary cloud subscriptions.

2. Critical Limitations & Regulatory Safeguards

1. Networked GSTIN Verification Fallback: Live automated verification requires active connectivity to the GST Common Portal API; in offline environments, the platform relies on strict local regex validation and mock provider adapters.

2. Pre-Printed Stationery Support: The WeasyPrint vector PDF engine is optimized for blank A4 digital paper; continuous-feed dot-matrix tractor stationery requires specialized plain-text esc/p driver extensions.

3. Future Roadmap and Research Extensions

- Government E-Invoicing & IRP Integration: Direct JSON payload dispatch to the Goods and Services Tax Network (GSTN) Invoice Registration Portal (IRP) for automated Invoice Reference Number (IRN) and QR-code generation.
- Automated E-Way Bill Generation: Extension to synthesize Part-A and Part-B E-Way bill slips for consignment shipments exceeding statutory monetary limits (Rs 50,000).
- Double-Entry General Ledger: Expansion into full double-entry accounting with accounts receivable, accounts payable, and trial balance generation.
- Multi-Branch Warehouse Inventory Tracking: Real-time stock decrement upon invoice finalization with automated re-order triggers.

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